**COUNT():**

```
mydb=# SELECT COUNT(*) AS total_employees FROM emp
loyees;
 total_employees
-----------------
               8
(1 row)


mydb=# SELECT COUNT(email) AS employees_with_email
 FROM employees;
 employees_with_email
----------------------
                    7
(1 row)


mydb=# SELECT COUNT(DISTINCT dept_id) AS unique_de
partments FROM employees;
 unique_departments
--------------------
                  3
(1 row)
```

**SUM():**

```
mydb=# SELECT SUM(salary) AS total_salaries FROM e
mployees;
 total_salaries
----------------
      530000.00
(1 row)


mydb=#
mydb=# SELECT d.dept_name, SUM(s.amount) AS total_
sales
mydb-# FROM sales s
mydb-# JOIN employees e ON s.employee_id = e.emp_i
d
mydb-# JOIN departments d ON e.dept_id = d.dept_id

mydb-# GROUP BY d.dept_name;
  dept_name  | total_sales
-------------+-------------
 Marketing   |      500.00
 Engineering |     6000.00
(2 rows)
```

**AVG():**

```
mydb=# SELECT AVG(salary) AS average_salary FROM e
mployees;
   average_salary
--------------------
 66250.000000000000
(1 row)


mydb=#
mydb=# SELECT dept_id, AVG(age) AS avg_age FROM em
ployees GROUP BY dept_id;
 dept_id |       avg_age
---------+----------------------
       2 | 35.5000000000000000
       3 | 31.0000000000000000
       1 | 39.0000000000000000
(3 rows)
```

**MAX() and MIN():**

```
mydb=# SELECT MAX(salary) AS highest_salary, MIN(s
alary) AS lowest_salary FROM employees;
 highest_salary | lowest_salary
----------------+---------------
       85000.00 |      48000.00
(1 row)


mydb=#
mydb=# SELECT dept_id, MAX(hire_date) AS latest_hi
re FROM employees GROUP BY dept_id;
 dept_id | latest_hire
---------+-------------
       2 | 2026-10-12
       3 | 2023-03-15
       1 | 2026-10-12
(3 rows)
```

**STRING_AGG():**

```
mydb=# SELECT dept_id, STRING_AGG(first_name, ', '
) AS employee_names
mydb-# FROM employees GROUP BY dept_id;
 dept_id |    employee_names
---------+----------------------
       2 | Sarah, Olga
       3 | Peter, Anna
       1 | John, Jane, Mike, Tom
(3 rows)


mydb=#
mydb=# SELECT dept_id, STRING_AGG(first_name, ', '
 ORDER BY first_name) AS ordered_names
mydb-# FROM employees GROUP BY dept_id;
 dept_id |     ordered_names
---------+----------------------
       1 | Jane, John, Mike, Tom
       2 | Olga, Sarah
       3 | Anna, Peter
(3 rows)


mydb=# SELECT dept_id, ARRAY_AGG(salary) AS salary
_array FROM employees GROUP BY dept_id;
 dept_id |             salary_array
---------+--------------------------------------
       2 | {55000.00,52000.00}
       3 | {48000.00,65000.00}
       1 | {75000.00,85000.00,80000.00,70000.00}
(3 rows)
```

```
mydb=# SELECT dept_id, ARRAY_AGG(first_name ORDER
BY last_name) AS employees
mydb-# FROM employees GROUP BY dept_id;
 dept_id |      employees
---------+----------------------
       1 | {Mike,Tom,Jane,John}
       2 | {Sarah,Olga}
       3 | {Peter,Anna}
(3 rows)
```

**Statistical Functions:**

```
mydb=# SELECT dept_id, STDDEV(salary) AS salary_st
d_dev, VARIANCE(salary) AS salary_variance
mydb-# FROM employees GROUP BY dept_id;
 dept_id |   salary_std_dev   |    salary_variance

---------+--------------------+------------------
-----
       2 |  2121.320343559643 |   4500000.00000000
0000
       3 | 12020.815280171308 | 144500000.00000000
0000
       1 |  6454.972243679028 |  41666666.66666666
6667
(3 rows)


mydb=#
mydb=# SELECT CORR(experience_years, salary) AS ex
perience_salary_correlation FROM employees;
 experience_salary_correlation
-------------------------------
           0.24363231334560298
(1 row)
```

**GROUP BY: basic and multiple columns:**

```
mydb=# SELECT category, COUNT(*) AS number_of_prod
ucts, SUM(price) AS total_value, AVG(price) AS ave
rage_price
mydb-# FROM products GROUP BY category;
  category   | number_of_products | total_value |
   average_price
-------------+--------------------+-------------+-
--------------------
 Electronics |                  2 |     1034.99 |
517.4950000000000000
 Stationery  |                  1 |        5.00 |
  5.0000000000000000
 Fruit       |                  2 |       70.50 |
 35.2500000000000000
(3 rows)


mydb=#
mydb=# SELECT EXTRACT(YEAR FROM sale_date) AS year
,
mydb-#        EXTRACT(QUARTER FROM sale_date) AS q
uarter,
mydb-#        COUNT(*) AS total_sales,
mydb-#        SUM(amount) AS total_revenue
mydb-# FROM sales
mydb-# GROUP BY EXTRACT(YEAR FROM sale_date), EXTR
ACT(QUARTER FROM sale_date)
mydb-# ORDER BY year, quarter;
 year | quarter | total_sales | total_revenue
------+---------+-------------+---------------
 2024 |       1 |           1 |       1500.00
 2024 |       2 |           2 |       3300.00
 2024 |       3 |           2 |       1700.00
(3 rows)
```

**GROUP BY with expressions:**

```
mydb=# SELECT
mydb-#   CASE
mydb-#     WHEN age < 30 THEN 'Under 30'
mydb-#     WHEN age BETWEEN 30 AND 50 THEN '30-50'

mydb-#     ELSE 'Over 50'
mydb-#   END AS age_group,
mydb-#   COUNT(*) AS employee_count,
mydb-#   AVG(salary) AS avg_salary
mydb-# FROM employees
mydb-# GROUP BY
mydb-#   CASE
mydb-#     WHEN age < 30 THEN 'Under 30'
mydb-#     WHEN age BETWEEN 30 AND 50 THEN '30-50'

mydb-#     ELSE 'Over 50'
mydb-#   END;
 age_group | employee_count |     avg_salary
-----------+----------------+--------------------
 Under 30  |              3 | 59333.333333333333
 Over 50   |              1 | 70000.000000000000
 30-50     |              4 | 70500.000000000000
(3 rows)
```

**HAVING:**

```
mydb=# SELECT dept_id, COUNT(*) AS employee_count
mydb-# FROM employees GROUP BY dept_id HAVING COUN
T(*) > 2;
 dept_id | employee_count
---------+----------------
       1 |              4
(1 row)


mydb=#
mydb=# SELECT category, AVG(price) AS avg_price, C
OUNT(*) AS product_count
mydb-# FROM products GROUP BY category
mydb-# HAVING AVG(price) > 30 AND COUNT(*) >= 2;
  category   |      avg_price       | product_coun
t
-------------+----------------------+-------------
--
 Electronics | 517.4950000000000000 |
2
 Fruit       |  35.2500000000000000 |
2
(2 rows)
```

**Window VS aggregate functions:**

```
mydb=# SELECT dept_id, AVG(salary) AS dept_avg_sal
ary FROM employees GROUP BY dept_id;
 dept_id |  dept_avg_salary
---------+--------------------
       2 | 53500.000000000000
       3 | 56500.000000000000
       1 | 77500.000000000000
(3 rows)
```

```
mydb=# SELECT emp_id, first_name, dept_id, salary,

mydb-#        AVG(salary) OVER (PARTITION BY dept_
id) AS dept_avg_salary
mydb-# FROM employees;
 emp_id | first_name | dept_id |  salary  |  dept_
avg_salary
--------+------------+---------+----------+-------
-------------
      1 | John       |       1 | 75000.00 | 77500.
000000000000
      2 | Jane       |       1 | 85000.00 | 77500.
000000000000
      3 | Mike       |       1 | 80000.00 | 77500.
000000000000
      5 | Tom        |       1 | 70000.00 | 77500.
000000000000
      6 | Olga       |       2 | 52000.00 | 53500.
000000000000
      4 | Sarah      |       2 | 55000.00 | 53500.
000000000000
      8 | Anna       |       3 | 65000.00 | 56500.
000000000000
      7 | Peter      |       3 | 48000.00 | 56500.
000000000000
(8 rows)
```

```
mydb=# SELECT SUM(bonus) AS total_bonus, AVG(bonus
) AS avg_bonus, COUNT(bonus) AS employees_with_bon
us
mydb-# FROM employees;
 total_bonus |       avg_bonus       | employees_w
ith_bonus
-------------+-----------------------+------------
----------
    26500.00 | 5300.0000000000000000 |
           5
(1 row)
```

**Handling NULLs:**

```
mydb=# SELECT COUNT(*) AS total_rows, COUNT(email)
 AS non_null_emails,
mydb-#        COUNT(*) - COUNT(email) AS null_emai
ls
mydb-# FROM employees;
 total_rows | non_null_emails | null_emails
------------+-----------------+-------------
          8 |               7 |           1
(1 row)
```

**Common Patterns:**

- **Multiple Aggregations:**

```
mydb=# SELECT dept_id, COUNT(*) AS employee_count,
 MIN(salary) AS min_salary,
mydb-#        MAX(salary) AS max_salary, AVG(salar
y) AS avg_salary, STDDEV(salary) AS salary_std_dev

mydb-# FROM employees GROUP BY dept_id ORDER BY av
g_salary DESC;
 dept_id | employee_count | min_salary | max_salar
y |     avg_salary     |   salary_std_dev
---------+----------------+------------+----------
--+--------------------+--------------------
       1 |              4 |   70000.00 |   85000.0
0 | 77500.000000000000 |  6454.972243679028
       3 |              2 |   48000.00 |   65000.0
0 | 56500.000000000000 | 12020.815280171308
       2 |              2 |   52000.00 |   55000.0
0 | 53500.000000000000 |  2121.320343559643
(3 rows)
```

- **Conditional Aggregation:**

```
mydb=# SELECT dept_id, COUNT(*) AS total_employees
,
mydb-#        COUNT(CASE WHEN salary > 60000 THEN
1 END) AS high_earners,
mydb-#        COUNT(CASE WHEN hire_date > '2023-01
-01' THEN 1 END) AS recent_hires
mydb-# FROM employees GROUP BY dept_id;
 dept_id | total_employees | high_earners | recent
_hires
---------+-----------------+--------------+-------
-------
       2 |               2 |            0 |
     2
       3 |               2 |            1 |
     1
       1 |               4 |            4 |
     3
(3 rows)
```

- **Percentage of Total:**

```
mydb=# SELECT dept_id, COUNT(*) AS dept_count,
mydb-#        ROUND(COUNT(*)::NUMERIC / (SELECT CO
UNT(*) FROM employees) * 100, 1) AS percentage
mydb-# FROM employees GROUP BY dept_id ORDER BY pe
rcentage DESC;
 dept_id | dept_count | percentage
---------+------------+------------
       1 |          4 |       50.0
       2 |          2 |       25.0
       3 |          2 |       25.0
(3 rows)
```